The assembly of the low load fluted panels was somewhat simpler than that for either of the other two tubular panel designs in that no fingered external doublers and no rivets were required. Otherwise the assembly was identical to that for the other tubular panels.

### 23.1.4 Fluted Single Sheet Panel Assembly

The fluted single sheet panels were formed using two sets of brake forming dies. The first set of dies developed an exaggerated sine wave configuration and the second set of dies formed the flutes and provided the final configuration. As a final brake operation, a standard brake was used to set the flat edge of the panel.

Figure 23-5 shows a completed single sheet test panel. The end members were not intended to represent flight hardware. They were designed for ease of fabrication and to permit testing of the panel section. The panel end plates shown in Figure 23-5 were machined from thick aluminum bar stock and grooved to receive the end of the corrugated panel as described in section 22.2.1.

The end plate assembly procedure consisted of bonding the corrugated sheet to the end plate. The bonding material was Hysol 5189/3538 which is a two part epoxy. The end plate and the epoxy were heated to 115$^{o}$F to improve flow characteristics, and epoxy was injected into the groove by use of a hypodermic. The corrugated sheet was manually inserted into the groove, starting at one end and progressively springing adjacent beads into the groove, and finally squaring the assembly on a horizontal surface plate with large angle blocks for the 24 hour room temperature cure cycle. The edge doublers were installed with the same epoxy as that used for the end plate attachment.